SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED
CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)
Assessment Tool: Application-Based Questions
Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal
A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:
<form action="/register" method="POST">
<input type="text" name="name">
<input type="email" name="email">
<button>Submit</button>
</form>
When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.
Questions:
1. Identify appropriate HTML5 semantic elements missing in the structure.
2. Modify the structure to make it standards-compliant.
3. Interpret the HTTP request generated during form submission.
4. Analyze the HTTP response cycle from server to browser.
5. Suggest improvements for usability and accessibility.
2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome
<p>About us
</div>

</body>
</html>

Questions:

1. Identify improper nesting and missing closing tags.
2. Analyze how different browsers may interpret this structure.
3. Identify missing semantic elements (e.g., header, section).
4. Provide a corrected W3C-compliant HTML structure.
5. Suggest techniques to ensure cross-browser compatibility.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

<form action="/submit" method="GET">
<input type="text" name="username">
<input type="password" name="password">
<button type="button">Login</button>
</form>

Questions:

1. Identify issues in the form configuration.
2. Analyze why the HTTP request is not triggered.
3. Interpret the role of GET method in this scenario.
4. Propose a corrected implementation.
5. Suggest improvements for secure data transmission.

Code of Conduct:

I T.Srikanthreddy(192365030) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.Srikanth

1. Online Symposium Portal

Issues: Missing semantic elements, labels, and proper form accessibility.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Online Symposium Registration</title>
</head>
<body>

   <header>
      <h1>Online Symposium Portal</h1>
      <p>Register for the college symposium</p>
   </header>

   <main>
      <section>
         <h2>Student Registration</h2>

         <form action="/register" method="POST">
            <label for="name">Student Name:</label>
            <input type="text" id="name" name="name" required>

            <br><br>

            <label for="email">Email Address:</label>
            <input type="email" id="email" name="email" required>

            <br><br>

            <button type="submit">Register</button>
         </form>
      </section>
   </main>

   <footer>
      <p>&copy; 2026 College Symposium</p>
   </footer>

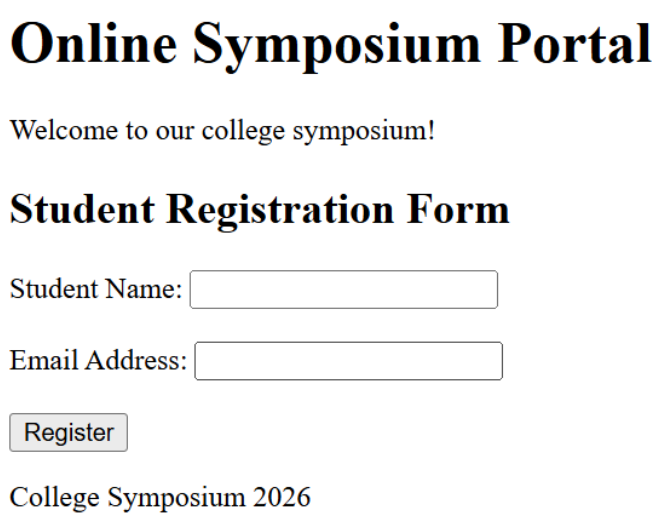
</body>
```

</html>

HTTP request and response

- Request: The browser sends a POST request to /register containing the submitted form data in the request body.
- Server processing: The server validates the name and email, processes the registration, and generates a response.
- Response: The server returns an HTTP status code, headers, and optionally a response body containing a confirmation page or error message.
- Browser: The browser displays the returned content.

OUTPUT:



2. Cross-Browser Compatibility Issue

Issues: The heading and paragraph are missing closing tags. The document also lacks a doctype, language attribute, metadata, and semantic elements.

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Company Website</title>
</head>
```

```
<body>

  <header>
    <h1>Welcome</h1>
    <nav>
      <a href="#about">About Us</a>
    </nav>
  </header>

  <main>
    <section id="about">
      <h2>About Us</h2>
      <p>Learn more about our company and services.</p>
    </section>
  </main>

  <footer>
    <p>&copy; 2026 Company Website</p>
  </footer>

</body>
</html>
```

Welcome

About Us

About Us

Learn more about our company and services.

© 2026 Company Website

3. Form Submission Failure

Issues: The button has type="button", which does not submit the form. Also, GET is unsuitable for transmitting passwords because the data can appear in the URL.

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Login Form</title>
</head>
<body>

   <main>
      <section>
         <h1>User Login</h1>

         <form action="/submit" method="POST">
            <label for="username">Username:</label>
            <input type="text" id="username"
                  name="username" required>

            <br><br>

            <label for="password">Password:</label>
            <input type="password" id="password"
                  name="password" required>

            <br><br>

            <button type="submit">Login</button>
         </form>
      </section>
   </main>

</body>
</html>
```

Quick revision table

Program	Main problem	Main correction
1. Symposium	Missing semantics and labels	Add semantic elements and accessible form fields

2. Cross-browser	Invalid/incomplete HTML	Close tags and use valid HTML5 structure
3. Login	Incorrect button type and unsafe method choice	Use type="submit", POST, and HTTPS

OUTPUT:

